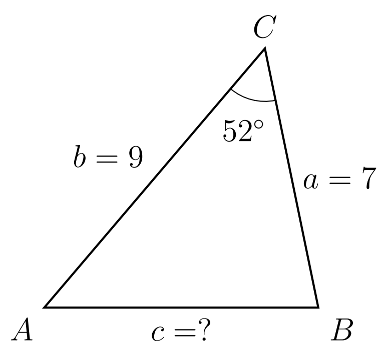
Solving Oblique Triangles with the Law of Cosines

Finding the third side of an SAS triangle, finding angles in an SSS triangle, and solving navigation problems with the law of cosines.

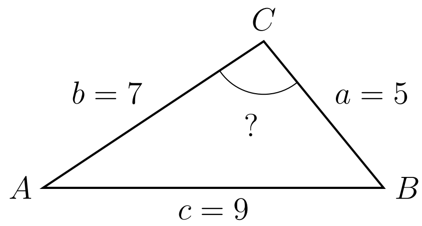
- Every triangle here is labelled the same way: side a is opposite vertex A, side b opposite B, side c opposite C. Each figure shows only that problem's own givens, with ? on what you are asked for.
- **Law of cosines:** $c^2 = a^2 + b^2 - 2ab\cos C$, and the same pattern for a and b. Rearranged for an angle: $\cos C = \dfrac{a^2 + b^2 - c^2}{2ab}$.
- Use it when you have **SAS** (two sides and the angle between them) or **SSS** (three sides). The law of sines cannot start either case.
- Keep full precision until the last line, then round sides to two decimal places and angles to two decimal places of a degree.
- Sanity-check every answer against the figure: the longest side must sit opposite the largest angle.

1. In triangle ABC, $a = 7$, $b = 9$, and the included angle $C = 52°$. Find the length of side c.



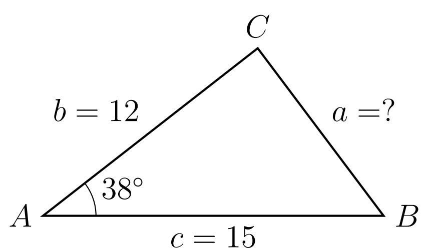
Answer: ______________

2. In triangle ABC the three sides are $a = 5$, $b = 7$, and $c = 9$. Find angle C.



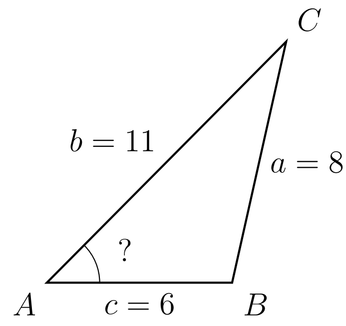
Answer: ____________________

3. In triangle ABC, $b = 12$, $c = 15$, and the included angle $A = 38°$. Find the length of side a.



Answer: ____________________

4. A triangular garden bed has sides $a = 8$, $b = 11$, and $c = 6$. Find angle A — the angle opposite the side of length 8.

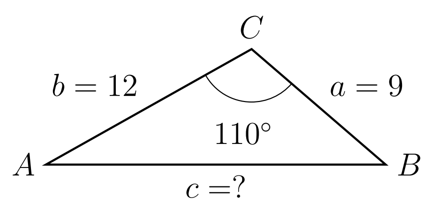


Answer: ________________

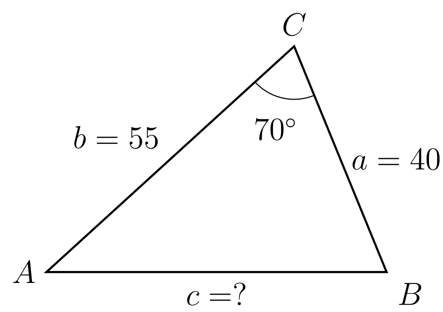
5. In triangle ABC, $a = 9$, $b = 12$, and $C = 110°$.

(a) Find side c: ________ (b) Find the area of the triangle: ________

For (b), the included angle is already known, so no new triangle solving is needed.



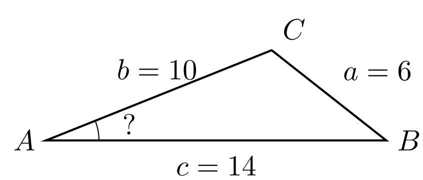
6. Two ships leave the same port at the same time. One sails 40 km on a bearing of 030°; the other sails 55 km on a bearing of 100°, so the angle between their courses at the port is 70°. How far apart are the ships?



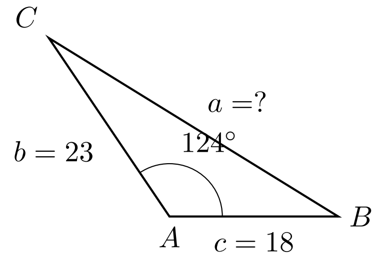
Answer: ________________ km

7. In triangle ABC, $a = 6$, $b = 10$, and $c = 14$.

(a) Write $\cos A$ as an exact fraction in lowest terms: ________ (b) Find angle A: ________

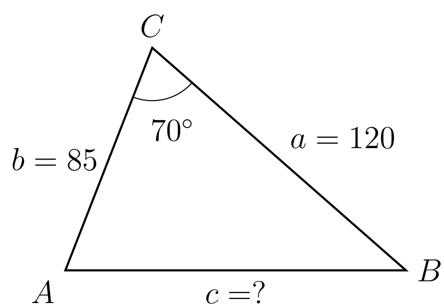


8. In triangle ABC, $b = 23$, $c = 18$, and the included angle $A = 124°$ is obtuse. Find side a.



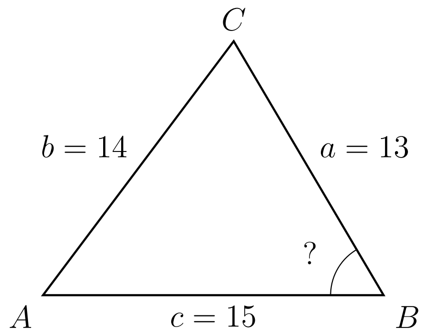
Answer: ________________

9. A pilot flies 120 miles from an airfield, then turns through a course change that leaves an interior angle of 70° at the turning point and flies a further 85 miles. How far is the plane from the airfield?



Answer: ________________ mi

10. A triangle has sides $a = 13$, $b = 14$, and $c = 15$. Find angle B.

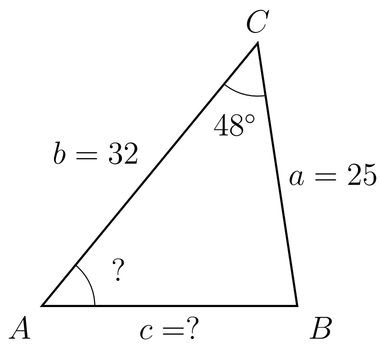


Answer: ____________________

11. In triangle ABC, $a = 25$, $b = 32$, and $C = 48°$.

(a) Find side c: __________ (b) Find angle A: __________

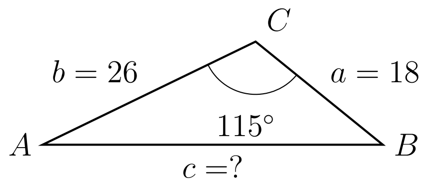
Once you have all three sides, you may use either law for part (b) — say which you chose and why.



12. A survey crew walks 18 km from marker A, turns, and walks 26 km to marker B. The interior angle at the turning point is $115°$.

(a) Find the straight-line distance from A to B.

(b) Explain why the law of sines cannot be used to start this problem, and what the law of cosines supplies that makes the law of sines usable afterwards.



Answer: ________________ km